Abstract Algebra - Homework 2

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Problem 1

Let $|G| = p^\alpha m$, where p is a prime, $\alpha \geq 1$ and $p \nmid m$. By definition of a p-Sylow subgroup, we have $|P| = p^\alpha$. Since $N \lhd G$, Lagrange's theorem implies $|N| \mid |G|$. Thus, we can write:

$$|N| = p^\beta d,$$

with $\beta \in [1, \alpha]$ and $d \mid m$, where $p \nmid d$. Consider the subgroup $N \cap P$. Since $N \cap P \leq P$, again applying Lagrange's theorem gives $|N \cap P| \mid |P| = p^\alpha$. Therefore, we must have:

$$|N \cap P| = p^\gamma,$$

where $\gamma \leq \beta$, hence $N \cap P$ must be a p-subgroup of N.

Consider the subgroup NP which contains N and P. Since $N \lhd G$ and $P \leq G$ we may apply the following formula,

$$|NP| = \frac{|N||P|}{|N \cap P|} = \frac{p^{\alpha+\beta}d}{p^\gamma} = p^{\alpha+\beta-\gamma}d$$

Since P is a p-Sylow subgroup of G it must also be a p-Sylow subgroup in any subgroup of G containing it. Therefore, its index in NP must be relatively prime to p.

$$[NP : P] = \frac{|NP|}{|P|} = \frac{p^{\alpha+\beta-\gamma}d}{p^\alpha} = p^{\beta-\gamma}d \implies p^{\beta-\gamma} = 1 \implies \beta = \gamma.$$

This shows $|N \cap P| = p^\beta$, which means it is a p-Sylow subgroup of N.

Problem 2

Problem 3

Problem 4

Problem 5

Problem 6